Much ado about Toric Codes

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1 The stabilizer formalism

1.1 "Minimal Working Example"

Consider a two bit Shor code (i.e., two qubit system) and the standard σ_z^i Pauli operators, where i indicates which qubit the operator will act on. What are the outcomes of applying $\sigma_z^1\sigma_z^2$ to the possible codes?

$$\begin{aligned}
\sigma_z^1\sigma_z^2 \left|00\right\rangle &= (+1 \times +1) \left|00\right\rangle = \left|00\right\rangle \\
\sigma_z^1\sigma_z^2 \left|11\right\rangle &= (-1 \times -1) \left|11\right\rangle = \left|11\right\rangle \\
\sigma_z^1\sigma_z^2 \left|10\right\rangle &= (-1 \times +1) \left|10\right\rangle = -\left|10\right\rangle \\
\sigma_z^1\sigma_z^2 \left|01\right\rangle &= (+1 \times -1) \left|01\right\rangle = -\left|01\right\rangle
\end{aligned}$$

Thus, we observe that $\sigma_z^1\sigma_z^2$ can pick up "bit flips" in the code. The same approach (though using σ_x^i Pauli operators) can also be used to pick up "phase" differences between codes; for example, let

$$\begin{aligned}
\psi &:= \left|000\right\rangle + \left|111\right\rangle \\
\phi &:= \left|000\right\rangle - \left|111\right\rangle
\end{aligned}$$

Then

$$\begin{aligned}
\sigma_x^1\sigma_x^2\sigma_x^3\psi &= \psi \\
\sigma_x^1\sigma_x^2\sigma_x^3\phi &= -\phi
\end{aligned}$$

In the above two examples, the operators $\sigma_z^1\sigma_z^2$ and $\sigma_x^1\sigma_x^2\sigma_x^3$ are said to *stabilize* the respective logical codes, i.e., the states corresponding to the encoding of logical 0 ($|00\rangle$,$|000\rangle$) or logical 1 ($|11\rangle$,$|111\rangle$) are eigenstates with eigenvalue $+1$. Note, these ideas generalize to larger and larger Shor codes[2] but in non-obvious ways (in terms of exactly which Pauli operators should be applied).

1.2 Stabilizer subgroup

Generalizing these examples, we're looking for members of a subgroup within the Pauli group G_k (on k qubits) called the *stabilizer* group (S_k). The defining property of S_k is that for any $g \in S_k \subset G_k$

$$g\,|0\rangle_k = +1\,|0\rangle_k$$
$$g\,|1\rangle_k = -1\,|1\rangle_k$$

where the subscript k indicates

$$|0\rangle_k := |\underbrace{00\cdots00}_{k}\rangle$$

A key property of the stabilizer group is that it is abelian, i.e., all of the members commute with each others. This enables us to simultaneously measure[1] all of the $g \in S_k$.

1.3 The code space

Okay, but this description kind of "puts the cart before the horse". We described how to detect errors in codes without really describing the encoding/code-space - we mention Shor codes but those actually aren't the only ones we can correct using this scheme. In general, the code space is the *linear subspace* T_{S_k} of all possible k-qubit states in the intersection of all the $+1$ eigenspaces of S_k

$$T_{S_k} := \left\{ |\psi\rangle \,\middle|\, g\,|\psi\rangle = +1\,|\psi\rangle \;\forall g \in S_k \right\}$$

That's a lot of fancy words but basically this means T_{S_k} consists of all sums

$$|\phi\rangle = \sum_k \alpha_k\,|\psi\rangle$$

where each $|\psi\rangle$ is **stabilized by all** $\forall g \in S_k$.

[1] If operators F and G commute, then you can come up with a "complete" set of basis states $|f_n g_n\rangle$ that are eigenstates of both F and G, and thus express any state of the quantum system as

$$|\Psi\rangle = \sum_n \alpha_n\,|f_n g_n\rangle$$

Measuring F you will get eigenstate f_m with probability $\left|\alpha_m^2\right|$ and the system will be in eigenstate $|f_m g_m\rangle$ and similarly for G. Thus, it doesn't matter which order you measure F, G in, you will get a consistent result. This is the significance of commuting operators.

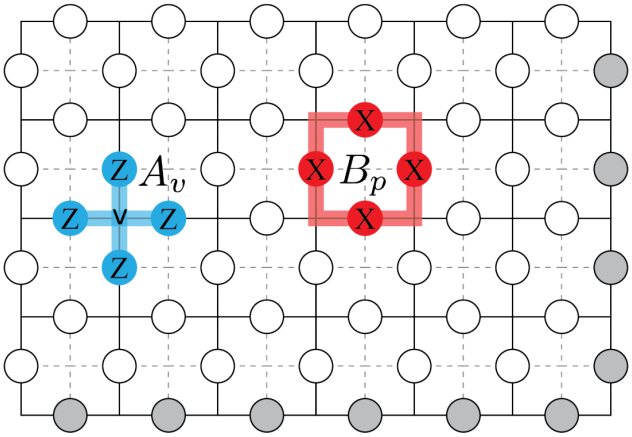


Figure 1: 2D Lattice with periodic boundary conditions with . v marks a particular vertex of the lattice and p marks a particular "face" of the lattice.

In general S_k is completely described by its l independent/unique *generators*[2] $\langle g_1, \ldots, g_l \rangle$. You can rigorously prove that the dimension[3] of T_{S_k} is equal to 2^{k-l}, i.e.,

$$|T_{S_k}| = 2^{k-l}$$

but intuitively you can think of it like this: for a k dimensional code space, without any constraints (i.e., just by default), you have 2^k basis states (independent, legal, k length qubit states). But then each stabilizer $g \in S_k$ constrains $|\psi\rangle$ to be in the $+1$-eigenspace of g and thereby cuts the possibilities by half. Hence after constraining with l stabilizers

$$2^k \times \underbrace{\frac{1}{2}\frac{1}{2}\cdots\frac{1}{2}}_{l} = 2^{k-l}$$

2 Vertex Operators and Plaquette Operators

Consider a 2D square lattice with periodic boundary conditions[4] and single qubit, spin-$\frac{1}{2}$ particles[5] on each edge (see figure 1). Take the standard $X := \sigma_x$ and $Z := \sigma_z$ Pauli operators. Assume the standard convention, that $|0\rangle, |1\rangle$are

[2] A generating set $\langle S \rangle$ of a group S is a subset of the group set such that every element of the group can be expressed as a combination (under the group operation) of finitely many elements of the subset and their inverses.

[3] I.e., dimension of the vector space of legal k length qubit states.

[4] i.e., a torus

[5] e.g., proton, neutron, electron, neutrino

eigenstates of Z, defined such that

$$Z|0\rangle = +1|0\rangle$$
$$Z|1\rangle = -1|1\rangle$$

Note that in the Z basis, the X operator acts as a "spin-flip" operator:

$$X|0\rangle = |1\rangle$$
$$X|1\rangle = |0\rangle$$

Then define "vertex" operators A_v and "plaquette" (face) operators B_p:

$$A_v := \prod_{j \in \text{star}(v)} Z_j$$
$$B_p := \prod_{j \in \text{body}(p)} X_j$$

where $\text{star}(v)$ is the set of particles identified with edges adjacent on the vertex v, and $\text{body}(p)$ is the set of particles identified with the edges surrounding the plaquette p. Note the product here is a tensor product. Then define the following Hamiltonian

$$H := -\sum_v A_v - \sum_p B_p \tag{1}$$

where $\sum_v$ is over all vertices in the lattice and $\sum_p$ is over all plaquettes. Recall the Hamiltonian of a system specifies its total energy.

Facts:

1. A_v and B_p are all Hermitian[6] and square to the identity[7]. Therefore they each (all) have eigenvalues of either $+1$ or -1.

2. Each of them commutes with all of the others

   (a) A_v commutes with $A_{v'}$ because they do not overlap at all

   (b) B_p and $B_{p'}$ overlap on at most one edge and X_j always commutes with $X_{j'}$ (and thus B_p and $B_{p'}$ commute)

   (c) If some A_v overlaps with some B_p then they overlap on two edges; the overlapping Z, X (one pair on each edge) each anti-commute but since there are two such pairs, the effect is cancelled out

3. Therefore, we can measure all the A_v and B_p simultaneously and the ground-state energy of H (i.e., the lowest energy state) will be the product of the $+1$ eigenstates of all of the A_v and B_p[8]

[6] H is Hermitian iff $\langle H\psi|\psi\rangle = \langle\psi|H\psi\rangle = \langle H^\dagger\psi|\psi\rangle$. Also sometimes called *self-adjoint*.

[7] $A_v \cdot A_v^\dagger \equiv I$

[8] Since, if you plug $A_v \equiv B_p \equiv 1$ into eqn. (1) you get the largest negative sum.

All of this should be looking familiar given section 1 and it should come as no surprise that $\{A_v, B_p\}$ (over all vertices and all faces) comprise a stabilizer subgroup (of the Pauli group G_k on the lattice, where k is the number of particles captured by the lattice). So what's the dimension of the code space T_{S_k} on this lattice with these stabilizers? Thinking a little bit imaginatively about $\{A_v, B_p\}$ as a set of constraints gives some physical meaning to them: violating any of the constraints (i.e., a state generating a measurement of -1 from any of the $\{A_v, B_p\}$) costs energy by virtue of the hamiltonian (any A_v, B_p that measures -1 will contribute a positive term to the sum in eqn. (1)). Thus, the ground-state of the system (i.e., the lowest energy state) should obey all of the "constraints" induced by $\{A_v, B_p\}$. All told the dimension of the code space is four [3].

3 Toric Codes

But what do the possible legal codes look like? Note (by the just preceding discussion on satisfying constraints), this is the same thing as considering what are the ground-states (lowest energy states) of the system. First, as a kind of warm up, consider only the A_v and the system state where each qubit is in the $|0\rangle$ state. In such a state, all A_v would be measured in the $+1$ state as well. What happens if we apply X to one or two qubits in a row (i.e., flip two spins)? In this case only vertex operators A_v adjacent to those qubits will be affected, through the Z_j that measures -1 on the flipped qubit (see figure 2 where in (a) we show how one such A_v is affected). Crucially, notice that for longer strings of errors (≥ 3 qubits) it's only the A_v at the ends of the string that will pick up the error. This ambiguity is what leads to techniques such as minimum weight perfect matching[4]. What happens if we flip a sequence of qubits that comprise a "loop" on the lattice? In that case, any A_v adjacent to any of the qubits will have 2 of its 4 Z_j measure -1 on the flipped qubits and but since

$$A_v := Z_{j_1} Z_{j_2} Z_{j_3} Z_{j_4} \equiv (+1 \times +1) \times (-1 \times -1) \equiv +1$$

A_v will still be measured in its $+1$ eigenstate. Indeed there are many, many such possible states (with some number of non-flipped qubits but also loops of flipped qubits). This is called *degeneracy*, when lots of distinct physical states are measured to have the same energy. In particular, if we only measure A_v, loops that go over the edge and back aroud (i.e., all the way around the torus), will also be indistringuishable from the groundstate.

Recall that what we're trying to accomplish in constructing error correcting codes is to discover particular states of our quantum system (in this case, a lattice of spin-$\frac{1}{2}$ particles) that can be distinguished between using non-destructive measurements (in this case, using stabilizers). Therefore such degeneracy is problematic. The resolution is to apply the B_p, i.e., measure B_p *as well as* the A_v. What does this accomplish? Well looking at figure 3, it seems like not much, but in fact we've gained a useful ability: we can use the B_p to completely eliminate any loop of spin flips, by applying the B_p operators adjacent

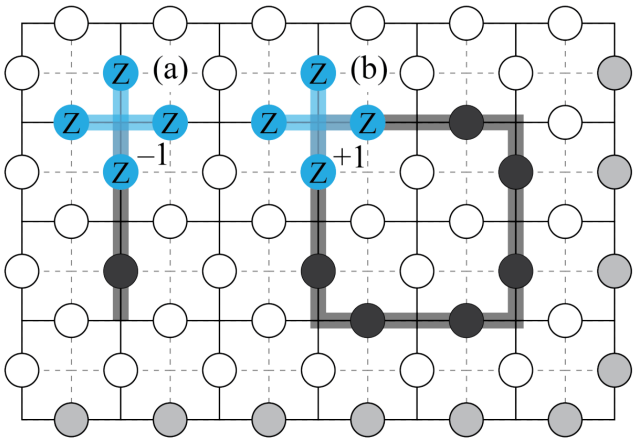


Figure 2: "Flipped qubits" are shaded black. In (a) we consider a short string of flipped qubits and in (b) we consider a loop of flipped qubits.

to the interior edge of the loop (see figure 4). But measuring B_p along with A_v doesn't resolve all ambiguities; consider a loop of flipped spins that extends all the way around the edge of the lattice (i.e., wraps around an axis of the torus). Applying B_p to each particle along such a loop would simply shift the loop laterally. **Thus, we have found the set of undetectable errors for this coding scheme**, i.e., the set $\{\bar{X}_1, \bar{X}_2, \bar{Z}_1, \bar{Z}_2\}$ of *Toric codes* (see figure 5). The fact that there are only four such codes is intimately related to the dimension[1] of the stabilizer group (which we earlier computed to be 4). Note, that the distance between codes is the width or height of the lattice (whichever is larger).

References

[1] Chris. Quantum codes and quantum error correction, 2018.

[2] Daniel Gottesman. Stabilizer codes and quantum error correction, 1997.

[3] Paul Herringer. The toric code, 2020.

[4] Oscar Higgott. Pymatching: A python package for decoding quantum codes with minimum-weight perfect matching, 2021.

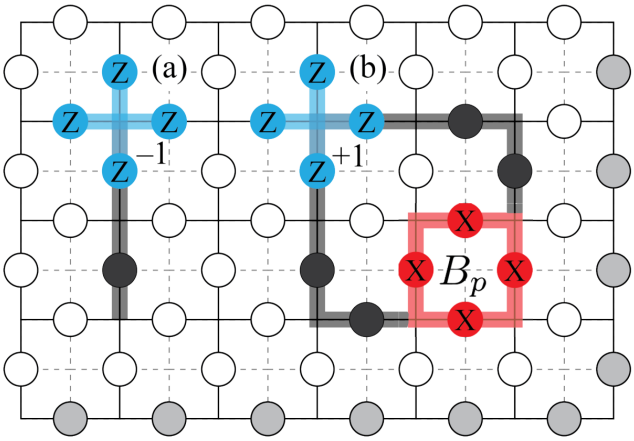


Figure 3: Both A_v and B_p being measured.

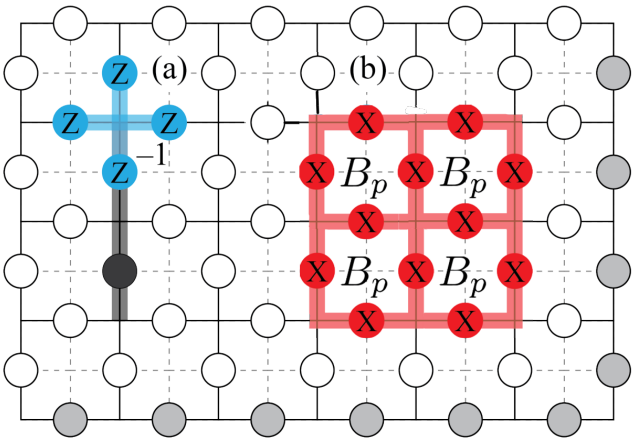


Figure 4: Using B_p to collapse the loop in fig. 3 - notice that the X operators flip the flipped spins "back to normal" *and* cancel each other out (on the interior).

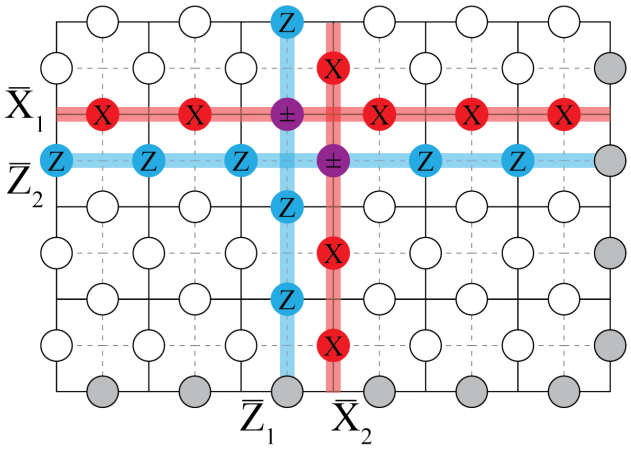


Figure 5: The fourse possible logical operators $\bar{X}_1$, $\bar{X}_2$, $\bar{Z}_1$, $\bar{Z}_2$.